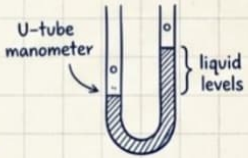
Solutions

A Complete Guide to Physical Chemistry



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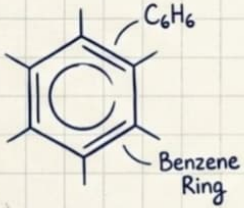


Concentrations: Part 1

Mass %

1) $$\boxed{w/w\ \% = \frac{\text{wt. of solute}}{\text{wt. of soln}} \times 100}$$

↳ How much solute is present in 100g of soln.

2) $$\boxed{w/v\ \% = \frac{\text{wt. of solute}}{\text{vol of soln in mL}} \times 100}$$

↳ How much wt is present in 100mL of soln.

3) $$\boxed{v/v\ \% = \frac{\text{vol of solute}}{\text{vol of soln}} \times 100\ \text{mL}}$$

Concentrations: Part 2

4) $$\boxed{\text{Strength} = \left(\frac{\text{wt. of solute}}{\text{vol of soln}}\right) \times 1000\ \text{mL}}$$

↳ How much wt is present in 1000 mL of soln.

5) $$\text{ppm} = \left(\frac{\text{wt. of solute}}{\text{vol of soln}}\right) \times 10^6 \quad (\text{i.e., } 10^4 \times \text{w/v \%})$$

6) $$\text{ppb} = 10^7 \times \text{w/v \%}$$

7) $$\text{ppt} = 10^{12} \times \text{w/v \%}$$

Expert Note 📌

The standard definitions relate to powers of 10. ppm = 10^6, ppb = 10^9, ppt = 10^12.

Since w/v% is already out of 10^2, multiplying it by 10^4 gives 10^6 (ppm), 10^7 gives 10^9 (ppb), and 10^10 gives 10^12 (ppt). The notes say 10^12 × w/v%, which would technically be 10^14. Just remember: ppt means 10^12 total!

* Molarity :- moles of solute in 1 L of soln.

for glucose :-

180g —— 1 mol	22.5g —— 0.125 mol
90g —— 0.5 mol	18g —— 0.1 mol
45g —— 0.25 mol	9g —— 0.05 mol

Q: What is the molarity if 90g of glucose is added to 500 mL of water?

90g of glucose in 1000mL → 0.5 mol

$\therefore$ 90g of glucose in 500mL → $\left[\frac{0.5}{1000}\right] \times 500 =$ **0.25 M**

Expert Note 📌
Watch out for this common student trap!
90g of glucose = 0.5 moles.
Molarity = moles / volume in L. So, 0.5 moles / 0.5 L (500mL) = 1.0 M.
The calculation written here accidentally calculated the moles in 250mL instead!

* Normality :-

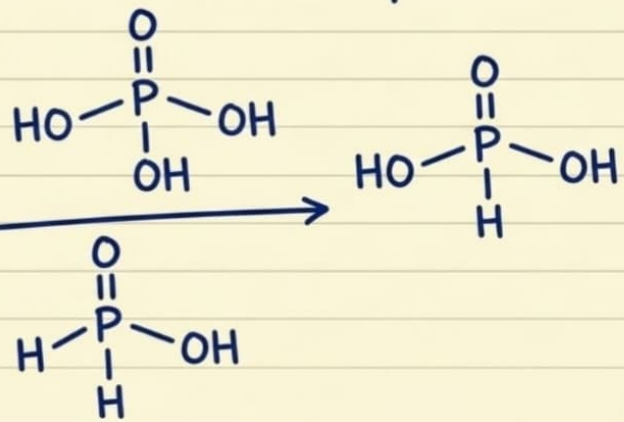
$$H_2SO_4 + NaOH \longrightarrow Na_2SO_4 + H_2O$$

$$H_3PO_4 + NaOH \longrightarrow NaH_2PO_4 + H_2O$$

$$HCl + NaOH \longrightarrow NaCl + H_2O$$

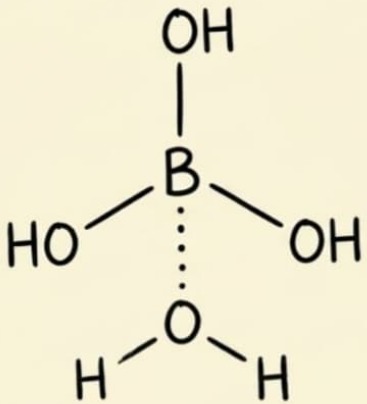
equivalents are the no. of H^+ or OH^- replaced.

- H_2SO_4 — 2 equiv.
- H_3PO_4 — 3 equiv.
- H_3PO_3 — 2 equiv.
- H_3PO_2 — 1 equiv.



* Normality (Continued)

- H_3BO_3 — ~~3 equiv.~~ → Actually 1 equiv!



[∴ It is a non-protic acid. It is basically Lewis acid as it has vacant orbital.]

* Bases/Salts Equivalents:

- $NaOH$ — OH^- — 1 eq
- $Mg(OH)_2$ — 2 eq
- Al_2O_3 — 6 eq

* Eq. wt. :-

- H_2SO_4 — 98g — 2 eq ∴ 1 eq = 49g
- H_3PO_4 — 98g — 3 eq ∴ 1 eq = 32.7g
- HCl — 36.5g — 1 eq

Ions:

Mg^{+2} — M/2
Al^{+3} — M/3
O^{-2} — M/2
Cl^{-1} — M/1

* Henry's Law :- (Solubility of solids in liquids)

Temp:

$\Delta H_{soln} > 0$ — Endo ; (Inc in temp, inc in solub)

$\Delta H_{soln} < 0$ — Exo ; (Dec in temp, inc in solub)

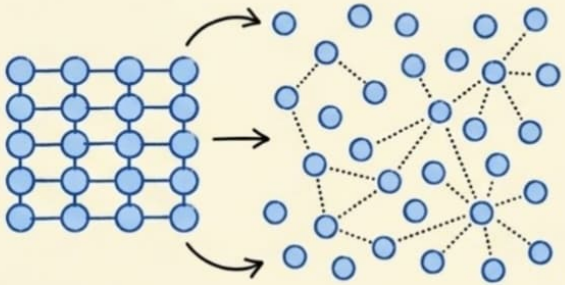
Pressure: No significant effect

* Energy Diagram:

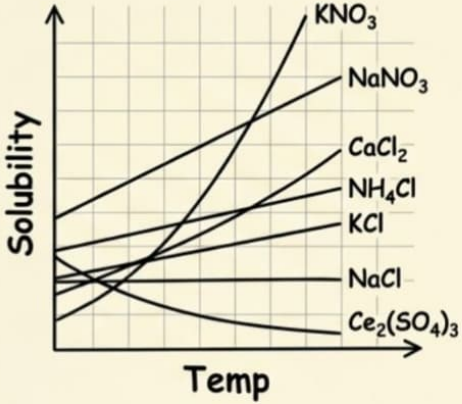
Solute + Solvent → Energy released / absorbed

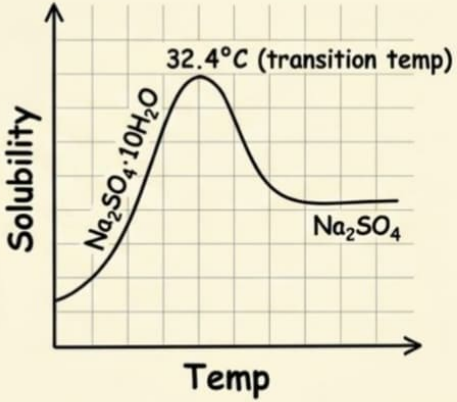
$CaCl_2 + aq \longrightarrow sol + \Delta$ (Exo – energy released)

$NH_4Cl + aq \longrightarrow sol - \Delta$ (Endo – energy absorbed)

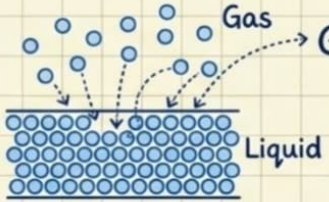


Note: Henry's Law applies to gases in liquids, not solids. The title is a common misnomer here.





* Solubility of gases in liquids :-



Generally, exo as $\Delta H < 0$

$\therefore$ T should be low & P should be high.

$\therefore M_p \propto P$

$\therefore$ Henry's law states that the solubility of gas in liquid is directly prop to its partial pressure above the liquid.

$$P = K_H \cdot X$$

(K_H = Henry's constant, depends upon the nature of gas)

e.g. at 25°C :
He — 144·97 kbar
N_2 — 76·48 kbar
CO_2 — 1·67 kbar

Higher K_H, lower solub, lower K_H, higher solub.

Applications of Henry's Law :-

1. Soft drinks like CO_2.
2. Scuba divers: mix. of O_2, He & N_2 is in the cylinders of scuba divers.
   (He - 11.7%, O_2 - 32.1%, N_2 - 56.2%)
3. Higher alt.: p is less, diff in blood. Suffer anoxia i.e. thinking is less.
4. Bends: Gases speedingly eject when scuba divers come at surface after going deep in ocean, due to which they suffer med prob. Thus they are put to hyperbaric chamber, where the gaseous press is higher than atm press, and then it is released slowly.

*More concepts: $p \propto X$, where p is partial pressure.
When T inc, K_H inc but X dec.

Note: Henry's Law applies to gases in liquids, not solids. The title is a common misnomer here.

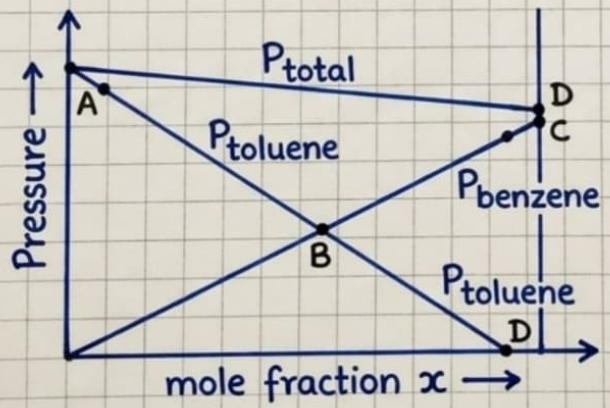
* Limitations :-

1. Ideal cond : very dilute which is possible when pressure is low & temp is high.
2. Gases must not react with solvent. Like H_2, He, N_2, O_2, CO_2, Ar, Kr, Xe, CH_4, HCHO.

Raoult's Law :-

- Vapour pressure : Vapour pressure is the pressure of the vapours when there is a dynamic equilib b/w rate of evaporation & rate of condensation.

Sol^n of liq in liq :



Here, $P_{total} = P_{benzene} + P_{toluene}$

$AD = BD + CD$

✳ State :-

$$P_{total} = P^{\circ}_{benzene} X_{benzene} + P^{\circ}_{toluene} X_{toluene}$$

∴ Partial vapour pressure of a component is proportional to the mole fraction of the compo.

$$P \propto X \quad \text{or} \quad P = k \cdot X$$

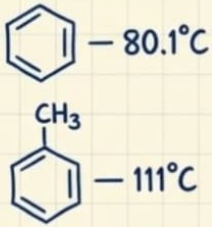
If $X = 1$ ∴ $P^{\circ} = k \cdot 1$ ∴ $P = P^{\circ} \cdot X$

Special case : Raoult's law is a spl case of Henry's law

$$P = K_H X \quad (\text{gas in liq})$$

$$P = P^{\circ} X \quad (\text{liq in liq})$$

✳ Konovaloff Rule :-



Intermolecular forces are more strong in Toluene as compared to benzene. ∴ Benzene is volatile, thus thus gas phase is richer with benzene vapours.

Expert Note: Konovaloff's Rule Rule predicts the composition of the vapor phase in equilibrium with a liquid mixture. The vapor is always richer in the more volatile component.

- **Dalton's law** (for gases only): $P_a = P_{total} \cdot y_a$

Ideal solutions :-

Solutions that obey Raoult's law.

- $\Delta H_{mix} = 0$ | $\Delta V_{mix} = 0$ | $W = 0, q = 0, \Delta E = 0$
- $\Delta G_{mix} = RT \sum X_i \ln X_i$
- $\Delta S_{mix} = -R \sum X_i \ln X_i$

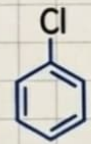
ΔH is 0 because the total amt of energy get consumed will get released as solute & solvent molecules get bonded.

These soln are mixtures of compounds having almost nearly same str.

Examples:

n-hexane

n-heptane



Chlorobenzene

Bromobenzene

C_2H_5Br Ethyl bromide

C_2H_5Cl Ethyl chloride

Low VP means High BP.

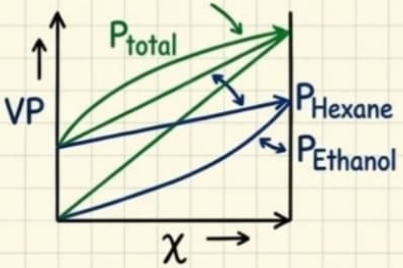
Expert Note:
Ideal solutions are theoretical constructs where solute-solute, solvent-solvent, and solute-solvent interactions are identical. Raoult's Law is obeyed at all concentrations and temperatures.

Non-Ideal Solutions: Deviations from Raoult's Law

Straight lines present ideal cond & curved lines present non-ideal cond.

Expert Note: Positive deviations (upward curve) show weaker A-B interactions than A-A or B-B, leading to higher VP and lower BP (azeotropes). Negative deviations (downward curve) show stronger A-B interactions, resulting in lower VP and higher BP (azeotropes).

+ve deviation :



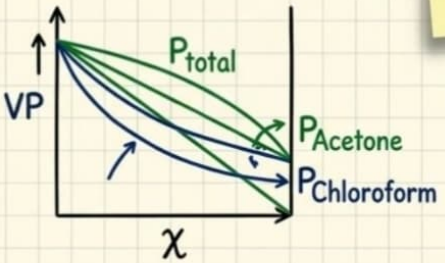
ΔH = +ve (Endothermic solns)

$\Delta H > 0$ because these molecules don't interact much among each other as they interacted among their own species.

Thus their VP inc & BP dec.

e.g., Hexane & Ethanol

-ve deviation :



ΔH = -ve (Exothermic solns)

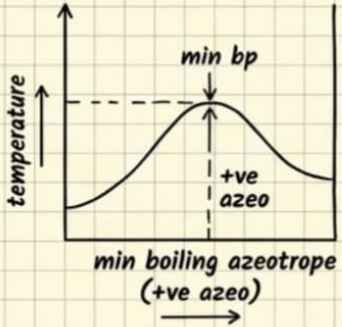
VP dec & BP inc due to strong interactions.

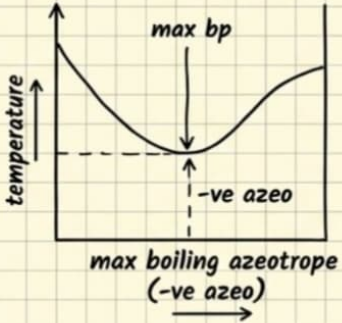
Interaction of A--B > A--A & B--B, thus $\Delta V < 0$.

e.g., Acetone & Chloroform

* Azeotropes :-

Azeo — boil, trope — turning.
They have no change in composition while boiling.
There are 2 types of azeotropes.





Expert Note: Positive azeotropes have a boiling point lower than either pure component, often due to repulsive interactions. Negative azeotropes have a boiling point higher than either pure component, typically due to strong intermolecular attractions like hydrogen bonding.

Ethanol & Water (+ve azeotropes) :-

Ethanol - 95.63%, 78.4°C | Water - 4.37%, 100°C

∴ The mix of ethanol will boil at 78.2°C which is lower than Ethanol & water.
(ΔV = +ve, ΔH = +ve)

HCl & Water (-ve azeo) :-

HCl - 20.2%, -84°C | Water - 79.8%, 100°C

∴ their mix has bp of 110°C which is greater than HCl & water.
(ΔV = -ve, ΔH = -ve)

Van't hoff factor (i) :-

Dissociation :

$$NaCl\,(aq) \longrightarrow Na^+ + Cl^-$$

bfr : 100 → 0,0 | aftr : 20 → 80,80.

∴ % diss = 80%

Generalised : $A^+B^- \rightleftharpoons A^+ + B^-$

Initial : 1 | After : $1-\alpha + 2\alpha$

e.g. $HClO \rightleftharpoons H^+ + ClO^-$

$$\therefore X_n \rightleftharpoons nX$$

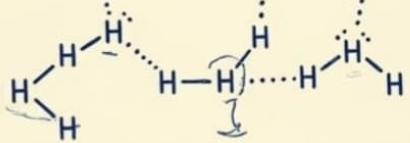
$$\Rightarrow \boxed{i_{diss} = 1 - \alpha + n\alpha}$$

Association (Reverse rxn) :

$$nX \rightleftharpoons X_n$$

Bfr : 1 → 0 | Aftr : $1-\alpha \rightarrow \frac{\alpha}{n}$

e.g. :- H—F ··· H—F ··· H—F



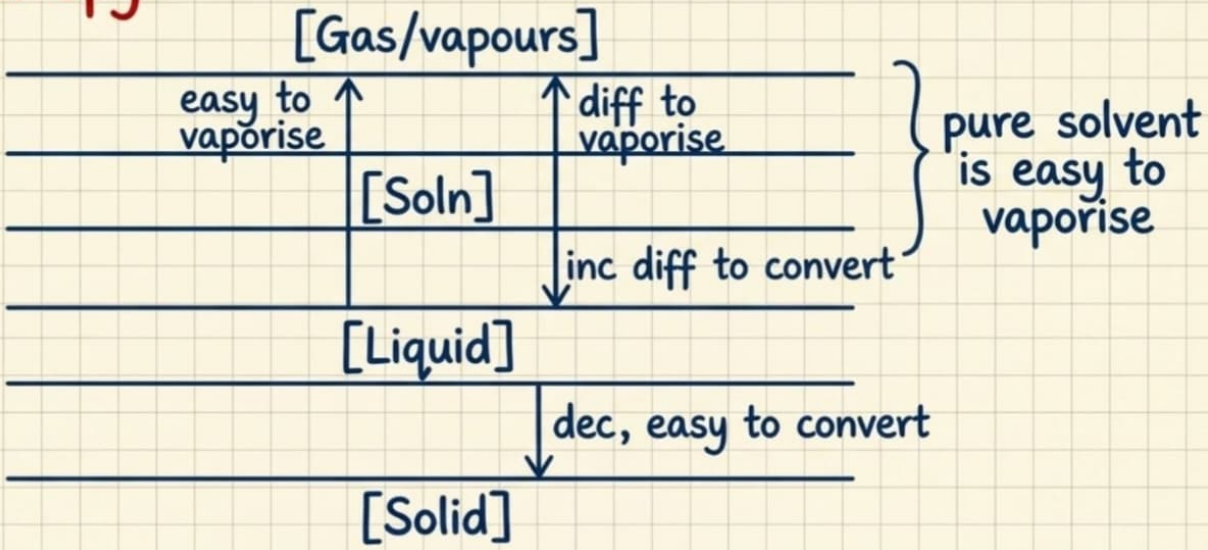
If n particles form 1 molecule and there are α such particles, then α/n = total molecules formed.

$$\boxed{\therefore\ i_{ass} = 1 - \alpha + \frac{\alpha}{n}}$$

Colligative properties :-

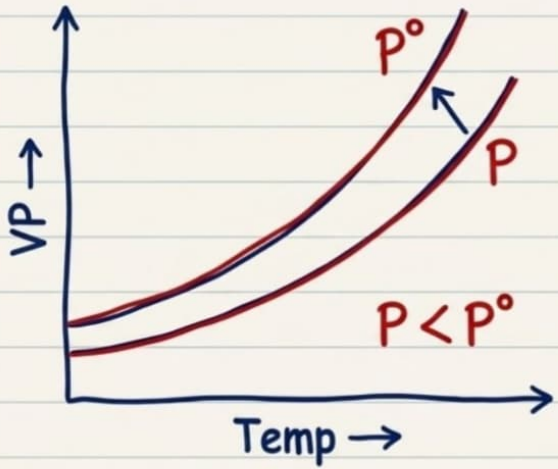
These prop depend upon no. of particles.

* Entropy :-



* Relative Lowering of Vapour Pressure

Pure solvent has more VP than soln. Since soln has already higher ΔS than pure solvent, diff of ΔS b/w soln & gas/vapour phase is also less & that of pure solvent & gas/vapour phase is more. Thus it gets easily converted to gas as compared to soln.



Derivation:

$$P < P^\circ \quad \therefore P = P^\circ \cdot X_A$$

$$\text{Since } X_A < 1 \quad \therefore P = P^\circ(1 - X_B)$$

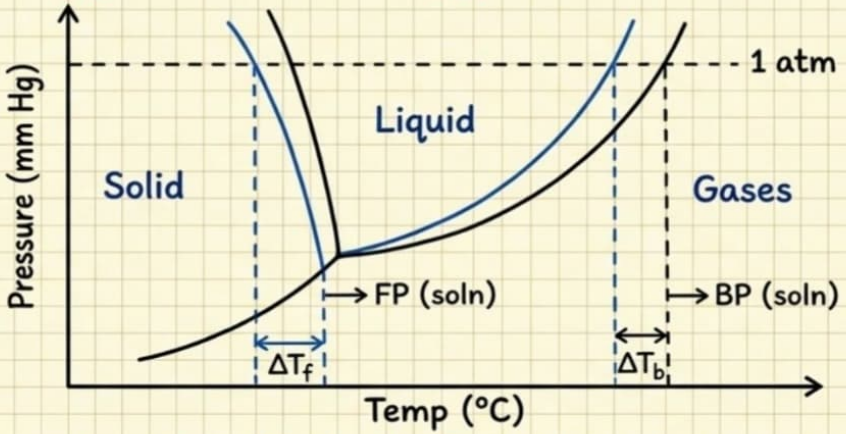
$$\text{or} \quad P = P^\circ - P^\circ X_B$$

$$\text{or} \quad X_B = \frac{P^\circ - P}{P^\circ}$$

$$\therefore \frac{P^\circ - P}{P^\circ} = \frac{W_b \times M_a}{M_b \times W_a}$$

$$\boxed{\therefore \frac{\Delta P}{P^\circ} = i X_b}$$

✳ Cryoscopy & Ebullioscopy :-



Blue line represents VP of soln & black line that of pure solvent. As we can see that the BP of soln is higher as compared to pure solvent & FP is lower.

✳ Boiling point (Ebullioscopy) :

On inc of 1 mol of glucose, there is an inc of 0.52°C in boiling pt.

$\Delta T_b \propto m$ or $\Delta T_b = i K_b m$

$$\Delta T_b = \frac{K_b \times W_b}{M_b \times W_a} = K_b \times m$$

K_b is called as Ebullioscopic constant.

Expert Note: The factor 'i' (Van't Hoff factor) accounts for dissociation or association of solute particles in solution, affecting ΔT_b. For non-electrolytes like glucose, i = 1.

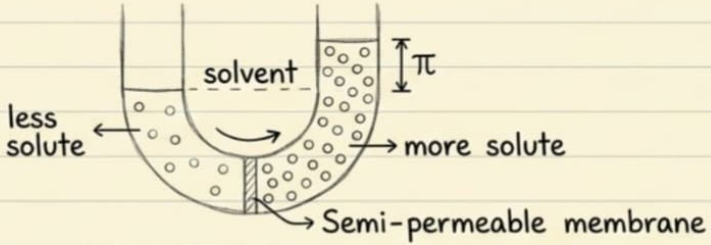
* Freezing Point Depression

$$\Delta T_f = i K_f \times m$$

Solvent	K_b	K_f
Benzene	2·53	5·12
Camphor	5·95	37·7
Chloroform	3·63	4·70
Dimethyl ether	2·02	1·79
Ethanol	1·22	1·99
Water	0·51	1·86

This data suggests that e.g. for water, 1 mol of water inc its bp by 0·51 K & fp by 1·86.

* Osmosis :-



✳ Osmotic pressure :-

$$\pi V = n_b RT \quad \text{or} \quad \pi = cRT$$

$$\therefore \pi V = \left(\frac{W_b}{M_b}\right) \times RT \quad \therefore \boxed{\pi = i\,cRT}$$

✳ Uses of CP (Tonicity) :-

1. Isotonic solns are those soln in whic ... ute
2. Hypertonic soln are those soln in whi
3. Hypotonic soln are those soln in whi ... ute

Expert Note 📌

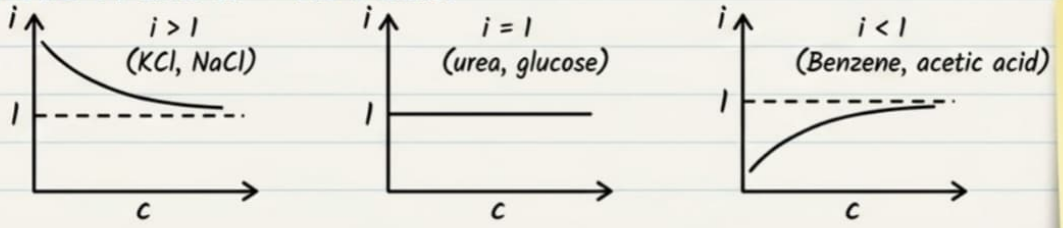
The shorthand here is a bit loose! Isotonic means two solutions have the *same* osmotic pressure. Hypertonic means a higher concentration of solute relative to another. Hypotonic means a lower concentration of solute.

✳ 2. Reverse osmosis :-

Sea water (highly conc) has high OP.

Abnormal Molecular masses :-

0.91% w/w soln of NaCl is used with RBC as they are isotonic in which NaCl is 0.15 M and π = 7.34 atm.



Expert Note: Isotonic solutions have the same osmotic pressure (π).

$$i = \frac{\text{Normal molar mass}}{\text{Abnormal mass}} = \frac{\text{Obs. C.P}}{\text{Calc. C.P}} = \frac{\text{Particle after ass/diss}}{\text{particle before.}}$$

The Case of Acetic Acid :

In benzene, benzoic acid dimerises & thus no. of molecules reduce to 1/2.

Benzene is non-polar but acetic acid is polar, thus ass occurs. ($P + P \rightarrow$ diss, $P + NP \rightarrow$ ass)

$$i_{CH_3COOH} = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}} = \frac{60}{120} = \frac{1}{2}.$$

Thus, CH_3COOH shows ass.